Reinforcement Learning in the Iterated Prisoner's Dilemma

Convergence and Comparative Analysis of Strategies

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The Prisoner's Dilemma

The Prisoner's Dilemma

One-shot game. Two players simultaneously choose to cooperate (C) or betray (B).

Payoffs satisfy $T > R > P > S$:

- $T = 4$ — temptation to defect
- $R = 3$ — mutual cooperation
- $P = 1$ — mutual defection
- $S = 0$ — sucker's payoff

Core tension. Betraying is strictly dominant, yet $(B, B) = (1, 1)$ is Pareto-dominated by $(C, C) = (3, 3)$.

	C	**B**
C	$(3, 3)$	$(0, 4)$
B	$(4, 0)$	$(1, 1)$

Payoff matrix (row = Player 1)

The Iterated Version (IPD)

- Repetition changes the analysis: defection can now be *punished*, cooperation *rewarded*.
- Axelrod (1984) showed that **Tit for Tat** — cooperate first, then mirror — wins computer tournaments against a wide strategy set.

Research question

Can a Q-learning agent learn an effective strategy in the IPD, and under what conditions does it converge to cooperation or defection?

Experimental Setup

Simulator Architecture & Strategy Set

Architecture.
All pairs played exhaustively for $T = 100{,}000$ rounds.

Naive strategies

- AlwaysCooperate / AlwaysBetray
- RandomAction ($p = 0.5$)
- ProbaCooperation (p configurable)

Reactive strategies

- TitForTat (TFT)
- SuspiciousTFT
- TitForTwoTats (lenient)
- TwoTitsForTat (punitive)
- Joss (TFT + noise)
- DiscriminatingAltruist
- Bully

The Q-Learning Model

Mathematical Formulation

State space. $s_t = (\text{own action}_{t-1},\ \text{opponent action}_{t-1})$.

Action space. $\mathcal{A} = \{C, B\}$.

Reward. r_t = payoff matrix value at round t.

Bellman update rule:

$$Q(s_t, a_t) \ \leftarrow\ Q(s_t, a_t) + \alpha\Big[r_t + \gamma \max_{a'} Q(s_{t+1}, a') - Q(s_t, a_t)\Big]$$

ε**-greedy policy.** Explore with prob. ε, exploit with prob. $1 - \varepsilon$.

ε decays multiplicatively: starts at 1, floors at 0.01.

Hyperparameter Sensitivity

Hyperparameter Sensitivity — Summary

ε — Exploration

Start at 1, decay to 0.01.
Too-fast decay $\Rightarrow$ premature exploitation of a suboptimal policy.
Too-slow $\Rightarrow$ still exploring at convergence.

Chosen: $\varepsilon_{\text{decay}} = 0.9999$.

α — Learning Rate

Too high: Q-values volatile, oscillating policy.
Too low: converges before learning against punitive strategies.

Chosen: $\alpha = 0.1$, robust across all opponents.

γ — Discount Factor

Most critical parameter.
Theoretical threshold $\gamma > \frac{1}{3}$ confirmed empirically.

Chosen: $\gamma = 0.9$.

Analogy: valuing future payoffs like a long-term asset.

Chosen hyperparameters:

$$\alpha = 0.1, \quad \gamma = 0.9, \quad \varepsilon_0 = 1.0, \quad \varepsilon_{\text{min}} = 0.01, \quad \varepsilon_{\text{decay}} = 0.9999$$

Key Finding — The γ Threshold

Since $T > R$ (betrayal always beats cooperation immediately), a myopic agent always defects.

Comparing discounted value streams against TitForTat:

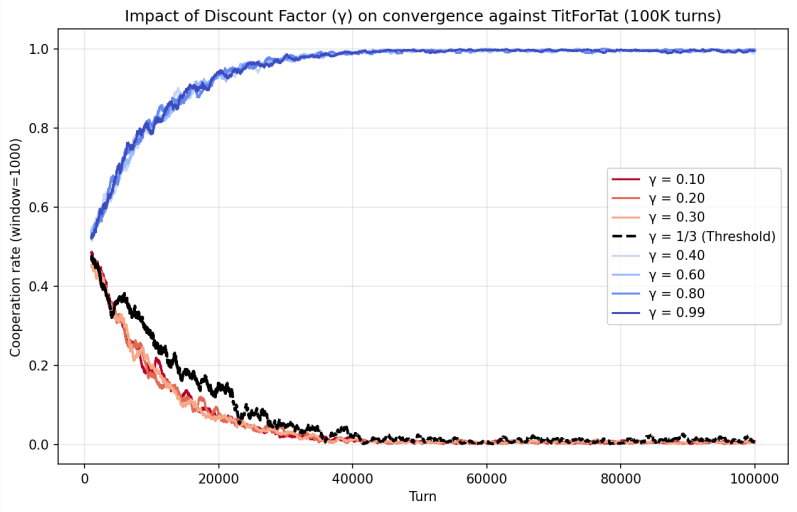
$$V_C = \frac{3}{1-\gamma}, \qquad V_B = 4 + \frac{\gamma}{1-\gamma}$$

Cooperation is rational iff $V_C > V_B$:

$$\frac{3}{1-\gamma} > 4 + \frac{\gamma}{1-\gamma} \implies \boxed{\gamma > \tfrac{1}{3}}$$

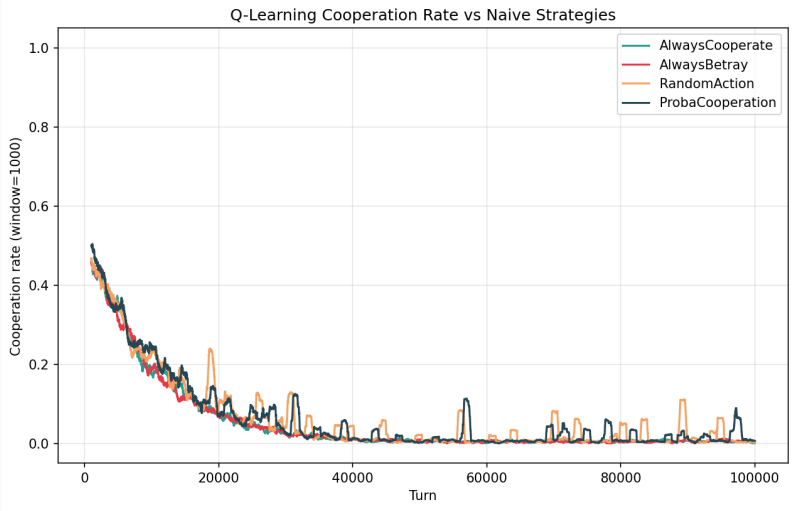
- This threshold is exactly recovered empirically in our simulations (Figure 5).
- The analogy to discounting future cash flows of a financial product is direct: cooperation is the long-term investment.
- Our chosen $\gamma = 0.9$ comfortably clears the threshold.

γ Threshold



Results

Against Naive Strategies



Against Naive Strategies

Strategy	Coop.	Exploit.	Vict.	Mut.C
AlwaysCooperate	1.4%	98.6%	0%	1.4%
AlwaysBetray	1.4%	0%	1.4%	0%
RandomAction	2.6%	48.7%	1.3%	1.3%
ProbaCoop. (0.7)	2.5%	68.2%	0.8%	1.8%

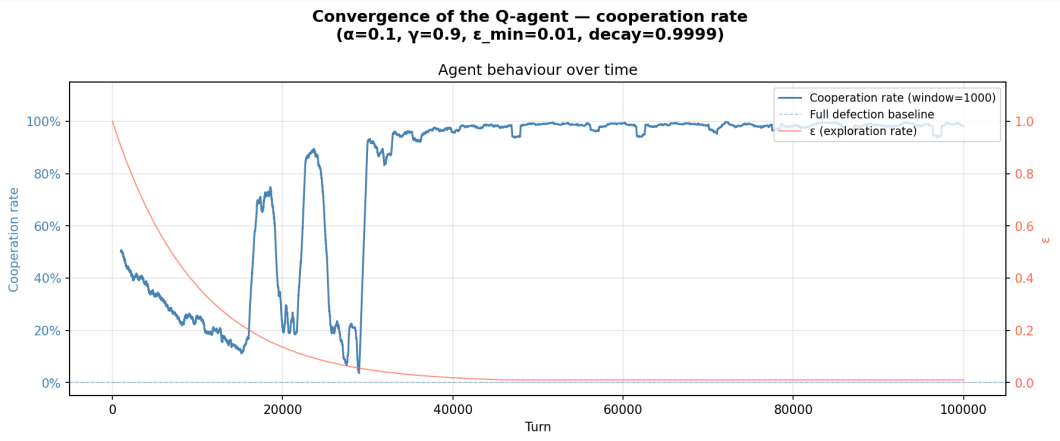
- Agent converges to systematic defection in all cases.
- Residual cooperation ($\sim$1%) is exploration noise, not learned behaviour.
- Exploitation rate tracks opponent cooperation probability closely.
- **Conclusion:** punishment of defection is necessary for cooperation to emerge.

Against Reactive Strategies

Strategy	Coop.	Exploit.	Vict.	Mut.C
TitForTat	98.6%	1.2%	1.2%	97.4%
SuspiciousTFT	98.6%	1.2%	1.2%	97.4%
TwoTitsForTat	98.5%	1.0%	2.2%	96.3%
Joss	98.4%	1.2%	11.0%	87.4%
TitForTwoTats	50%	49.8%	0.7%	49.3%

- Cooperation emerges against **TFT and variants**.
- Against **TitForTwoTats**: agent exploits leniency, defecting every other round — 50% cooperation, only 0.7% victimised.
- **Joss**: stochastic defection exploits the converged policy (11% victimisation).

Two Q-Learning Agents — Spontaneous Cooperation



Two Q-Learning Agents — Spontaneous Cooperation

- Both agents start from zero knowledge.
- Initial phase: mutual betrayal (high ε, temporary reward from defection).
- As ε decays, both agents sharply converge to mutual cooperation.
- Cooperation rate $\to 1$ consistently across all simulation runs.
- Convergence takes $\approx 30{,}000$ turns — comparable to convergence against pre-programmed strategies, underlining the role of exploration decay.

Why this matters. Cooperation is not programmed — it *emerges* from self-interest and repeated interaction alone.

Q-Learners in a Hostile Population

Setup. 10 agents: 7 AlwaysBetray + 3 QLearningStrategy. 100 000 rounds per pair.

Strategy	Avg. Score/Round
QLearningStrategy	**924 653**
AlwaysBetray	908 616

- Q-learners outperform the defector majority despite being a 30% minority.
- Against the 7 defectors: learn to defect back ($P = 1$ per round).
- Against each other: converge to cooperation ($R = 3$ per round).
- Even 30% Q-learning penetration is sufficient to generate outperformance.

Conclusion

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Key takeaways

- Against non-punishing opponents, Q-learning converges to systematic defection — consistent with Nash equilibrium.
- Against retaliatory strategies (TFT and variants), cooperation may arise, provided $\gamma > \frac{1}{3}$.
- Two independent Q-agents discover mutual cooperation from scratch.
- A minority of Q-agents outperforms majority in a hostile population.

Limitations. Memory depth of 1 limits recognition of longer-horizon strategies. Non-stationarity when two Q-agents face each other breaks convergence guarantees. Purely self-interested reward — no other-regarding preferences.

References i

- Robert Axelrod. *The Evolution of Cooperation*. Basic Books, 1984.
- Steven Kuhn. Prisoner's dilemma. *Stanford Encyclopedia of Philosophy*, 2019.
- Albert W. Tucker. A two-person dilemma. Mimeograph, Stanford University, 1950.